Basin Percolation in the LSR Associative Memory

1 Setup: LSR energy and the hard wall

The Log-Sum-ReLU (LSR) associative memory with M patterns $\boldsymbol{\xi}^1, \ldots, \boldsymbol{\xi}^M$ on the sphere $S^{N-1}(\sqrt{N})$ has energy

$$E(\mathbf{S}) = -\frac{N}{b} \ln \sum_{\mu=1}^{M} \big[\, 1 - b + b\, \varphi_\mu(\mathbf{S}) \,\big]_+, \qquad b \geq 2 + \sqrt{2}, \tag{1}$$

where $\varphi_\mu(\mathbf{S}) = (\boldsymbol{\xi}^\mu \cdot \mathbf{S})/N$ is the overlap and $[\cdot]_+ = \max(0, \cdot)$. The LSR-optimal value $b = 2 + \sqrt{2}$ maximizes capacity among $b = O(1)$ choices; Section 3.5 explores the regime $b \gg 1$.

The hard wall activates at

$$\varphi_c = \frac{b-1}{b} \ : \tag{2}$$

pattern μ contributes to the energy only when $\varphi_\mu(\mathbf{S}) > \varphi_c$. When no pattern is active, the argument of the logarithm vanishes and $E = +\infty$. For the optimal $b = 2+\sqrt{2}$ one obtains $\varphi_c = 1/\sqrt{2} \approx 0.707$.

Basin of a single pattern. For an isolated target $\boldsymbol{\xi}^1$, the accessible region is the spherical cap

$$X_0 = \big\{\, \mathbf{S} \in S^{N-1}(\sqrt{N}) \ : \ \varphi_1(\mathbf{S}) > \varphi_c \,\big\}, \tag{3}$$

with an infinite energy barrier at the boundary $\varphi_1 = \varphi_c$. The state is permanently confined to X_0.

2 Joint basins and continuous connectivity

With M patterns, the accessible region is the set where $\sum_\mu [\cdots]_+ > 0$, i.e. where at least one pattern is active. Two individual basins X_μ and X_ν form a *joint basin* if their intersection $X_\mu \cap X_\nu \neq \varnothing$.

Along any continuous path through $X_\mu \cap X_\nu$, both terms are simultaneously positive, so the energy is always finite—the state never encounters $\ln 0$. The energy does increase along the path, reaching a maximum at the "bottleneck" where the geodesic from $\boldsymbol{\xi}^\mu$ to $\boldsymbol{\xi}^\nu$ crosses the boundary of basin X_ν. At this saddle point, pattern ν's term has just activated ($\varphi_\nu = \varphi_c$, contributing zero) while the overlap with pattern μ is $\varphi_\mu^* = q\,\varphi_c + \sqrt{(1-q^2)(1-\varphi_c^2)}$. The barrier height is

$$\Delta E \ = \ \frac{N}{b} \ln \frac{1}{b(\varphi_\mu^* - \varphi_c)}\,, \qquad \varphi_\mu^* = q\,\varphi_c + \sqrt{(1-q^2)(1-\varphi_c^2)}\,, \tag{4}$$

where $q = (\boldsymbol{\xi}^\mu \cdot \boldsymbol{\xi}^\nu)/N$ is the mutual overlap. The barrier diverges at the *geometric threshold* $q_{\text{geom}} = 2\varphi_c^2 - 1$ (where the two caps just touch and $\varphi_\mu^* = \varphi_c$) and is finite for all $q > q_{\text{geom}}$.

Geometric overlap criterion. Two spherical caps of angular radius $\theta_c = \arccos \varphi_c$ overlap when the angle between their centers satisfies $\theta < 2\theta_c$, i.e. when

$$q = \frac{\boldsymbol{\xi}^\mu \cdot \boldsymbol{\xi}^\nu}{N} > \cos(2\theta_c) = 2\varphi_c^2 - 1 \;\equiv\; q_{\text{geom}}. \tag{5}$$

For $b = 2 + \sqrt{2}$, $\theta_c = \pi/4$ and $q_{\text{geom}} = 0$, so approximately $M/2$ random patterns satisfy $q > 0$ geometrically. However, for small $q > 0$ the barrier (4) reduces to $\Delta E \approx (N/b)\ln(\sqrt{2}/(bq))$, which scales as $\sim (N/2b)\ln N$ for $q \sim 1/\sqrt{N}$. For $b = O(1)$ this diverges as $O(N \ln N)$, making such corridors thermally inaccessible. For $b = \beta N$ the same formula gives $\Delta E \sim (\ln N)/(2\beta)$, which still diverges but only logarithmically, and may be thermally accessible at finite temperature (see Section 3.5). For general $b > 2 + \sqrt{2}$, the geometric threshold $q_{\text{geom}} > 0$ is positive, and the barrier diverges as $q \to q_{\text{geom}}^+$.

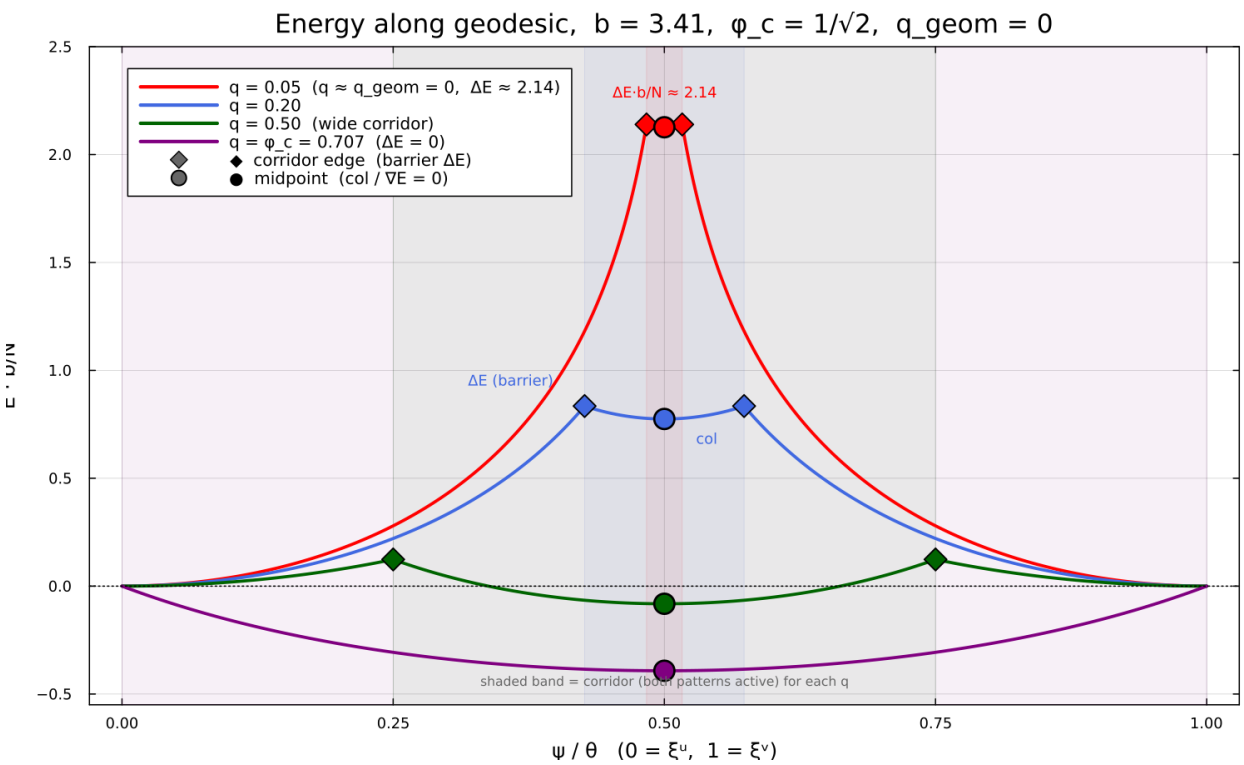

Figure 1: Energy $E \cdot b/N$ along the geodesic from basin centre $\boldsymbol{\xi}^\mu$ (left, $\psi/\theta = 0$) to basin centre $\boldsymbol{\xi}^\nu$ (right, $\psi/\theta = 1$), for four values of the mutual overlap $q = \boldsymbol{\xi}^\mu \cdot \boldsymbol{\xi}^\nu / N$ ($b = 2 + \sqrt{2}$, $\varphi_c = 1/\sqrt{2}$). The shaded region marks the *corridor* where both patterns are active simultaneously. **Diamonds (◇)** mark the corridor edges, where one pattern just activates ($\varphi_\nu = \varphi_c$): these are the energy maxima along the path and set the Monte Carlo barrier ΔE from Eq. (4). **Circles (•)** mark the midpoint $\psi = \theta/2$, which is a true critical point ($\nabla E = 0$ by symmetry) — the *col* or mountain pass — and lies *below* the corridor edges. As $q \to \varphi_c \approx 0.707$ (purple curve) the corridor widens to its maximum and the barrier vanishes; as $q \to 0$ (red curve, narrow corridor) the barrier grows as $(N/b)\ln(\sqrt{2}/(bq))$.

3 Percolation framework

3.1 Thermal connectivity threshold

We define two basins as *thermally connected* if the energy barrier along the optimal path through their overlap is comparable to the thermal energy scale T. This requires a minimum mutual overlap $q > q_{\text{eff}}(T)$.

At zero temperature, only basins with a wide, low-barrier corridor are connected. The natural hard-wall threshold is $q_{\text{eff}} = \varphi_c$, since patterns with $q > \varphi_c$ are already active at the target center (the term $[1 - b + b\,q]_+ > 0$).

3.2 Poisson model for the number of neighbors

For random patterns on $S^{N-1}(\sqrt{N})$, the overlap $q = (\boldsymbol{\xi}^\mu \cdot \boldsymbol{\xi}^1)/N$ with a fixed target is approximately $\mathcal{N}(0, 1/N)$ for large N. The number of patterns with $q > \varphi_c$ follows a Poisson distribution with rate

$$\lambda = M \cdot \Pr(q > \varphi_c) \;\approx\; \frac{\exp\bigl(N(\alpha - \varphi_c^2/2)\bigr)}{\varphi_c\sqrt{2\pi N}}, \tag{6}$$

where $M = e^{N\alpha}$ and we used the Mill's ratio approximation for the Gaussian tail.

Finite-N correction. If $\boldsymbol{\xi}$ is drawn uniformly on $S^{N-1}(\sqrt{N})$, the overlap $q = (\boldsymbol{\xi} \cdot \boldsymbol{\xi}^1)/N$ is the projection of a uniform spherical point onto a fixed axis, rescaled by $\sqrt{N}$. The marginal density of such a projection is the $(N-1)$-dimensional area element of the sphere at latitude q, which gives the exact density

$$f(q) = \frac{1}{B\bigl(\frac{1}{2}, \frac{N-1}{2}\bigr)}\,(1 - q^2)^{(N-3)/2}, \qquad q \in (-1, 1), \tag{7}$$

where B is the beta function (equivalently, $(q+1)/2 \sim \text{Beta}(\frac{1}{2}, \frac{N-1}{2})$).

Comparison with the Gaussian tail. The Mill's ratio approximation in (6) treats q as $\mathcal{N}(0, 1/N)$, matching the first two moments of $f(q)$. However, the tails differ. Write $(1 - q^2)^{(N-3)/2} = \exp\bigl[\frac{N-3}{2}\ln(1 - q^2)\bigr]$ and expand the logarithm:

$$\frac{N-3}{2}\,\ln(1 - q^2) = -\frac{N-3}{2}\,q^2 \;-\; \frac{N-3}{4}\,q^4 \;-\; \cdots \tag{8}$$

The leading term $-\frac{N-3}{2}\,q^2$ differs from the Gaussian exponent $-\frac{N}{2}\,q^2$ by $+\frac{3}{2}q^2$, which is $O(1)$ and independent of N. This constant shift is absorbed into the normalization: the prefactor $1/B(\frac{1}{2}, \frac{N-1}{2})$ of the exact density (7) differs from the Gaussian prefactor $\sqrt{N/(2\pi)}$ precisely to compensate, so that both distributions integrate to unity. The next term $-\frac{N-3}{4}\,q^4$, however, provides *extra suppression* that is $O(N)$ at fixed q and *cannot* be absorbed into the normalization. At the threshold $q = \varphi_c$, this correction is $-\frac{N-3}{4}\,\varphi_c^4$, so the exact tail probability is smaller than the Gaussian tail by a factor $\sim e^{-(N/4)\varphi_c^4}$. For $b = 2 + \sqrt{2}$ ($\varphi_c = 1/\sqrt{2}$), this gives $\sim e^{-N/16}$.

The Mill's ratio therefore *overestimates* λ; the true percolation threshold is shifted to higher α for finite N. The empirical correction $\langle K\rangle \cdot e^{cN}$ with $c \approx 0.083$ (Fig. 3, Panel 2) collapses the exact-sphere data onto the analytical curves, consistent with the $e^{-N/16} \approx e^{-0.0625\,N}$ estimate (the remaining discrepancy in c comes from higher-order terms in (8) and subleading prefactor corrections). As $N \to \infty$, the relative correction $q^4/q^2 = q^2$ becomes negligible for the relevant $q \sim 1/\sqrt{N}$ regime, and the Gaussian approximation becomes exact.

3.3 Percolation transition

The basins of M random patterns form a random geometric graph on S^{N-1}: each basin is a node, and two nodes are connected if their mutual overlap exceeds q_{eff}. The connected component containing the target undergoes a **percolation transition** at $\lambda = 1$.

Setting $\lambda = 1$ in (6) with $q_{\text{eff}} = \varphi_c$:

$$\boxed{\alpha_c = \frac{\varphi_c^2}{2} + \frac{\ln(\varphi_c\sqrt{2\pi N})}{N} \xrightarrow{N\to\infty} \frac{\varphi_c^2}{2} = \frac{1}{2}\Big(1 - \frac{1}{b}\Big)^2 .} \tag{9}$$

This coincides exactly with the zero-temperature critical capacity $\alpha_{\text{th}} = \frac{1}{2}\varphi_c^2$ from the mean-field theory. For the optimal $b = 2 + \sqrt{2}$, $\alpha_c \to 1/4$.

Physical picture.

- **Subcritical** ($\alpha < \alpha_c$): $\lambda < 1$; the target's connected cluster has $O(1)$ patterns. The state is confined near the target $\to$ retrieval.
- **Critical** ($\alpha \approx \alpha_c$): $\lambda \approx 1$; cluster size diverges; critical fluctuations.
- **Supercritical** ($\alpha > \alpha_c$): $\lambda > 1$; a giant connected component forms. The state delocalizes among exponentially many basins $\to$ paramagnetic phase.

3.4 Temperature dependence

At temperature $T > 0$, thermal fluctuations lower the effective barrier, so narrower corridors become traversable: $q_{\text{eff}}(T) < \varphi_c$. The percolation threshold shifts to

$$\alpha_c(T) \approx \frac{q_{\text{eff}}(T)^2}{2}, \tag{10}$$

which decreases with T—the retrieval region shrinks, consistent with the phase diagram.

From the mean-field free energy calculation: $\alpha_c(T) = \frac{1}{2}[1 - f_{\text{ret}}(T)]^2$, where $f_{\text{ret}}(T) = u(\varphi(T)) - T\,s(\varphi(T))$ is the retrieval free energy. Identifying $q_{\text{eff}}(T) = 1 - f_{\text{ret}}(T)$ connects the percolation threshold to the thermodynamic phase boundary.

3.5 Dependence on the nonlinearity parameter b

The preceding sections hold for any $b \geq 2 + \sqrt{2}$, with $\varphi_c = (b-1)/b$ and $q_{\text{geom}} = 2\varphi_c^2 - 1$. The LSR-optimal value $b = 2 + \sqrt{2}$ is special: $\varphi_c = 1/\sqrt{2}$ and the "accidental" identity $\cos(2\arccos\varphi_c) = \cos(\pi/2) = 0$ gives $q_{\text{geom}} = 0$, so approximately $M/2$ random patterns satisfy the geometric overlap criterion, and the infinite energy barrier at $q = 0$ is the sole mechanism confining the state. For any $b > 2 + \sqrt{2}$, $q_{\text{geom}} > 0$ and fewer patterns overlap geometrically.

It is instructive to ask what happens when b is taken large, in particular $b = \beta N$ with $\beta > 0$ fixed. Three qualitative changes occur simultaneously.

1. Barriers become $O(1)$. The barrier (4) has prefactor N/b, which for $b = \beta N$ becomes $1/\beta$:

$$\Delta E \;=\; \frac{1}{\beta}\,\ln\frac{1}{\beta N\,(\varphi_\mu^* - \varphi_c)}\,. \tag{11}$$

For geometrically overlapping pairs, $\varphi_\mu^* - \varphi_c \approx (q - q_{\text{geom}})/(2\varphi_c)$ is $O(1/N)$, so $\beta N(\varphi_\mu^* - \varphi_c) = O(1)$ and the barrier is $O(1/\beta)$. In contrast to the $b = O(1)$ case where $\Delta E = O(N)$, here thermal fluctuations of order T can overcome the barrier for all geometrically overlapping basins. The distinction between geometric and thermal connectivity disappears: $q_{\text{eff}}(T) \approx q_{\text{geom}}$ for any $T > 0$.

2. Basins shrink to microscopic caps. The hard-wall threshold becomes

$$\varphi_c = 1 - \frac{1}{\beta N}, \tag{12}$$

so the angular radius of each basin is $\theta_c = \arccos \varphi_c \approx \sqrt{2/(\beta N)} \to 0$. The geometric overlap of two caps now requires

$$q > 2\varphi_c^2 - 1 = 1 - \frac{4}{\beta N} + \frac{2}{\beta^2 N^2} \approx 1 - \frac{4}{\beta N}. \tag{13}$$

For random patterns with $q \sim \mathcal{N}(0, 1/N)$, this requires q to be within $O(1/N)$ of 1—a $\sqrt{N}$-sigma event with probability $\sim e^{-N/2}$, far out in the Gaussian tail.

3. Percolation threshold shifts to $\alpha_c = 1/2$. The Poisson rate becomes

$$\lambda \;\approx\; \frac{e^{N\alpha}}{\varphi_c\sqrt{2\pi N}}\, e^{-N\varphi_c^2/2}. \tag{14}$$

Since $\varphi_c^2 = (1 - 1/(\beta N))^2 \to 1$ as $N \to \infty$, setting $\lambda = 1$ gives

$$\alpha_c \;\to\; \frac{\varphi_c^2}{2} \;\to\; \frac{1}{2} \qquad (b = \beta N \to \infty). \tag{15}$$

This is exactly the critical capacity of the exponential (Modern Hopfield) associative memory $E \propto -\ln \sum_\mu e^{N\varphi_\mu}$, which corresponds to the formal limit $b \to \infty$ of the LSR energy (1).

Physical interpretation. The two limits represent qualitatively different physics:

	$b = O(1)$ (LSR)	$b = \beta N$ (exp. AM)
Barrier height	$O(N)$, hard confinement	$O(1)$, easy hopping
Cap angular radius	$O(1)$	$O(1/\sqrt{N})$
Geometric neighbors	$\sim M/2$ ($q > 0$)	exponentially rare
Percolation threshold	$\alpha_c = \frac{1}{4}$	$\alpha_c = \frac{1}{2}$
Confinement mechanism	energy barriers	geometric isolation
Temperature role	lowers q_{eff}, shrinks retrieval	negligible (barriers already $O(1)$)

In the $b = O(1)$ regime, basins are wide but tall barriers select which connections are thermally accessible—the phase boundary depends on T via $q_{\text{eff}}(T)$. In the $b = O(N)$ regime, basins are narrow and geometrically isolated; barriers are irrelevant because they are $O(1)$, and the phase boundary is temperature-independent at leading order.

Interpolation. For intermediate $b = N^\gamma$ with $0 < \gamma < 1$, the barrier scales as $N^{1-\gamma}$ and the cap radius as $N^{-\gamma/2}$. The crossover between the two regimes occurs at $\gamma = 1$, where barriers become $O(1)$ and confinement switches from energetic to geometric. The percolation threshold interpolates smoothly:

$$\alpha_c \;=\; \frac{\varphi_c^2}{2} \;=\; \frac{1}{2}\Big(1 - \frac{1}{b}\Big)^2 \;\xrightarrow{b\to\infty}\; \frac{1}{2}. \tag{16}$$

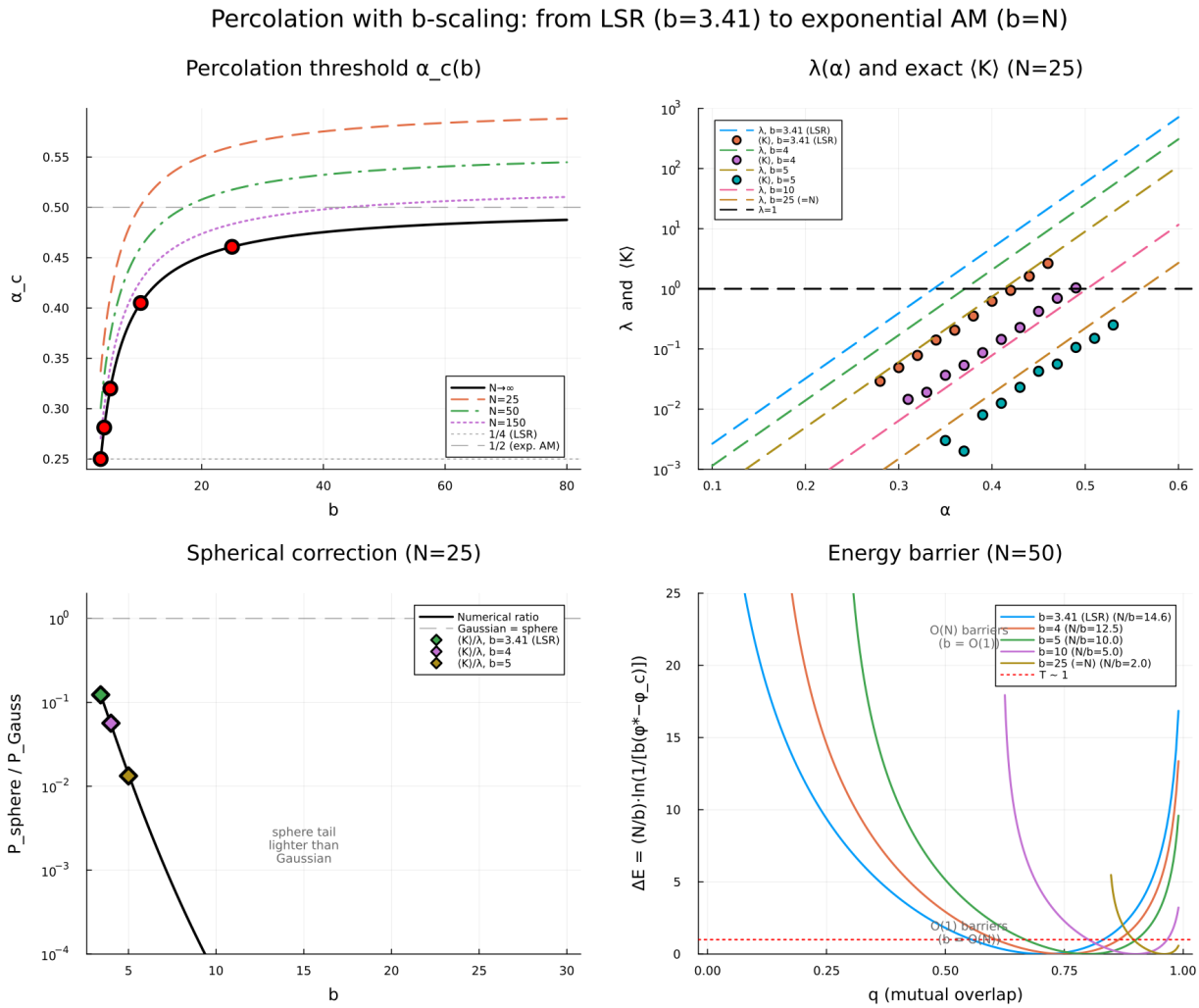

Figure 2: **Percolation with b-scaling: from LSR ($b = 2+\sqrt{2}$) to exponential AM ($b = N$).** **Panel 1** (top-left): Percolation threshold $\alpha_c(b)$ from (16) at $N \to \infty$ (solid), $N = 25$ (dashed), 50 (dash-dotted), 150 (dotted). Red dots mark the tested values $b \in \{3.41, 4, 5, 10, 25\}$. The threshold interpolates from $\alpha_c = 1/4$ at $b = 2+\sqrt{2}$ (LSR) toward $\alpha_c = 1/2$ (exponential AM, $b \to \infty$), with finite-N corrections moving α_c upward. **Panel 2** (top-right): Gaussian-approximate neighbor count $\lambda(\alpha)$ (dashed) and exact $\langle K \rangle$ (circles) for $b = 3.41, 4, 5$ at $N = 25$. For each b, the data points fall below the corresponding λ curve by a factor that grows with b (equivalently, with φ_c), reflecting the increasingly large finite-N spherical correction. Analytical curves for $b = 10$ and $b = 25$ are shown without data: at these thresholds ($\varphi_c = 0.9$ and 0.96), the exact spherical cap probability is so small that $\langle K \rangle = 0$ in all Monte Carlo trials. **Panel 3** (bottom-left): Spherical correction factor $P_{\text{sphere}}/P_{\text{Gauss}}$ vs b at $N = 25$. The solid curve is obtained by numerically integrating the exact density $f(q) \propto (1 - q^2)^{(N-3)/2}$ and dividing by the Gaussian tail $\bar{\Phi}(\varphi_c\sqrt{N})$. Diamond markers show the empirical ratio $\langle K \rangle/\lambda$ averaged over all tested α values. The agreement confirms that the Gaussian approximation overestimates the cap probability by a factor that depends on b (via φ_c) but is approximately constant in α. **Panel 4** (bottom-right): Energy barrier $\Delta E = \frac{N}{b} \ln\big(1/\big[b(\varphi_\mu^* - \varphi_c)\big]\big)$ from (4), plotted vs. the mutual overlap q at $N = 50$. For $b = O(1)$ (LSR), barriers are $O(N)$ and confine the state even at moderate temperature. For $b = \beta N$, barriers collapse to $O(1)$ (below the red dashed line $T \sim 1$), and thermal fluctuations easily overcome them—confinement becomes purely geometric.

4 BFS depth: chain connections

Starting from the target, we explore the basin connectivity graph via breadth-first search:

1. **Depth 0**: target pattern $\boldsymbol{\xi}^1$ with basin X_0.
2. **Depth 1**: all patterns μ with $(\boldsymbol{\xi}^\mu \cdot \boldsymbol{\xi}^1)/N > \varphi_c$ ($K_1 \sim \text{Poisson}(\lambda)$ patterns).
3. **Depth 2**: for each depth-1 pattern μ, find patterns ν with $(\boldsymbol{\xi}^\mu \cdot \boldsymbol{\xi}^\nu)/N > \varphi_c$ that are *not* already at depth 1.

Depth-2 patterns are absent for large N. For a depth-1 pattern μ (overlap $\varphi_\mu > \varphi_c$ with target) and a candidate depth-2 pattern ν (overlap $\varphi_\nu < \varphi_c$ with target), their mutual overlap is

$$\frac{\boldsymbol{\xi}^\mu \cdot \boldsymbol{\xi}^\nu}{N} \approx \varphi_\mu \cdot \varphi_\nu + \sqrt{(1-\varphi_\mu^2)(1-\varphi_\nu^2)}\,\frac{u_\mu \cdot u_\nu}{\sqrt{N-1}}, \tag{17}$$

where u_μ, u_ν are random unit vectors in the $(N-1)$-dimensional subspace perpendicular to the target.

The deterministic part satisfies $\varphi_\mu \cdot \varphi_\nu < \varphi_c^2 < \varphi_c$ (since $\varphi_c < 1$), and the random fluctuation $\sim \mathcal{O}(1/\sqrt{N})$ cannot bridge the gap $\varphi_c - \varphi_c^2 = \varphi_c(1-\varphi_c)$ for large N.

At finite N, the fluctuation has standard deviation $\sim \sqrt{(1-\varphi_c^2)(1-\varphi_\nu^2)}/\sqrt{N-1}$, requiring a $\sim \varphi_c(1-\varphi_c)\sqrt{N}$ sigma event for a depth-2 connection. For $b = 2+\sqrt{2}$ ($\varphi_c = 1/\sqrt{2}$), the gap is ≈ 0.207 and the requirement is $\sim 0.207\sqrt{N}$ sigma: for $N = 25$: $\sim 1\,\sigma$ (rare but possible); for $N = 50$: $\sim 1.5\,\sigma$ (very rare); for $N \geq 100$: negligible.

5 Simulation design

The accompanying script `percolation_LSR.jl` tests four aspects of the percolation hypothesis at the optimal $b = 2+\sqrt{2}$ ($\varphi_c = 1/\sqrt{2}$):

1. **Panel 1: Analytical** $\lambda(\alpha)$ for $N = 30, 50, 75, 150$. All curves cross $\lambda = 1$ near $\alpha \approx 0.25$, with finite-N corrections $\sim \ln N/N$ shifting the threshold to higher α.
2. **Panel 2: Direct neighbor counting** ($N = 25$, 3000 realizations). Generate $M = e^{N\alpha}$ random patterns on $S^{N-1}(\sqrt{N})$, count neighbors K with overlap $> \varphi_c$. Compare histogram with Poisson(λ). The Mill's ratio approximation overestimates λ at moderate N because the exact spherical tail is lighter than the Gaussian.
3. **Panel 3: Branching process survival** ($N = 50$). Simulate a Galton–Watson process with Poisson(λ) offspring. The survival probability transitions from 0 to 1 at α_c, sharpening with BFS depth (analogous to increasing system size in percolation).
4. **Panel 4: BFS depth analysis** ($N = 25$, 200 realizations). Count depth-1 neighbors K_1 and *new* depth-2 neighbors K_2. Confirms that $\langle K_2 \rangle \ll \langle K_1 \rangle$: the BFS effectively terminates at depth 1.

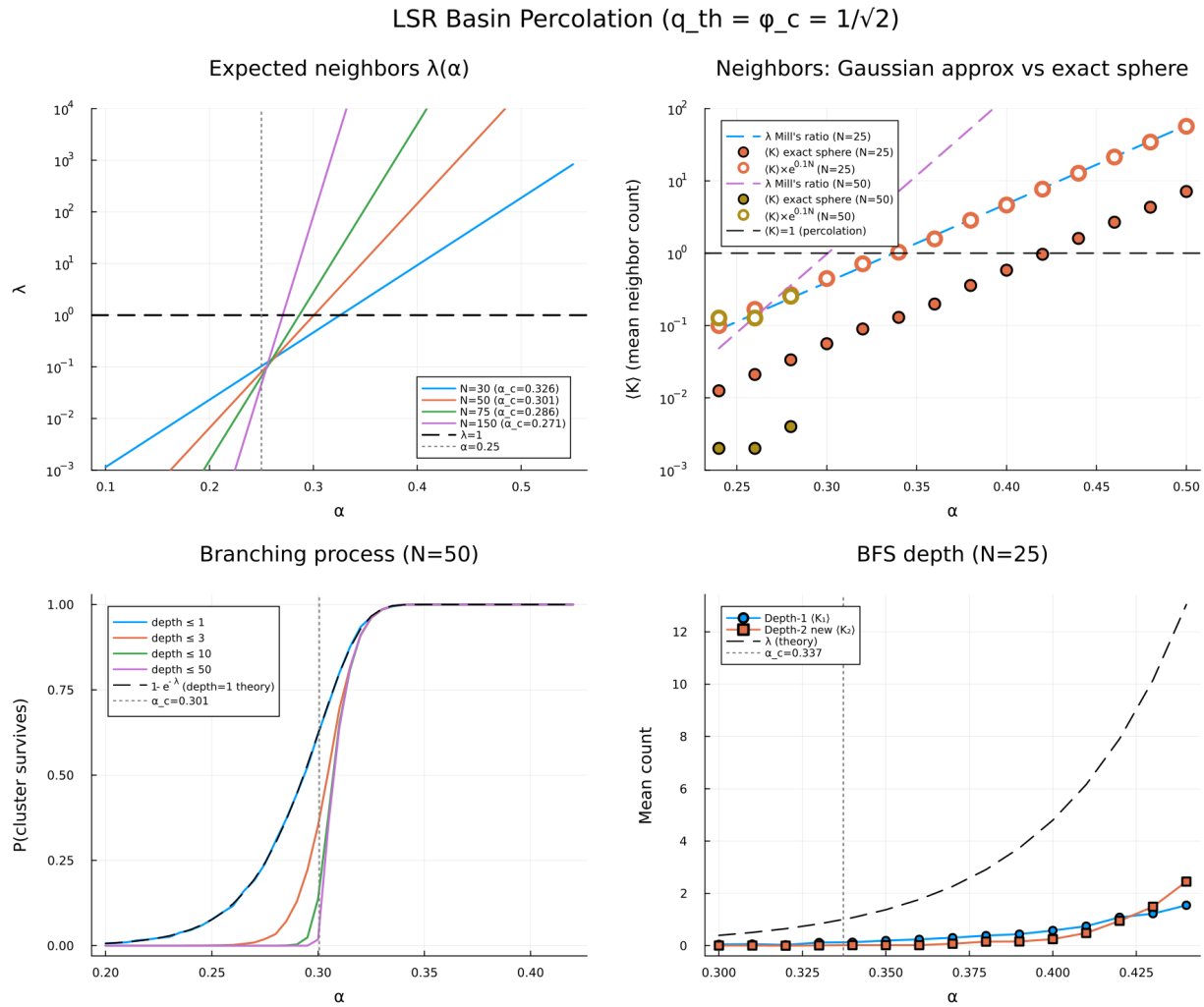

Figure 3: **Basin percolation in the LSR model. Panel 1** (top-left): Analytical expected neighbor count $\lambda(\alpha)$ from the Mill's ratio approximation (6) for $N = 30, 50, 75, 150$. All curves cross the percolation threshold $\lambda = 1$ (dashed line) near $\alpha \approx 0.25$; the finite-N shift $\alpha_c(N) - \alpha_\infty \sim \ln N/N$ is clearly visible. **Panel 2** (top-right): Comparison of the Gaussian (Mill's ratio) prediction for λ (dashed curves) with the mean neighbor count $\langle K \rangle$ obtained by exact enumeration on $S^{N-1}(\sqrt{N})$ (filled circles; $N = 25, 50$). The systematic downward shift of $\langle K \rangle$ relative to λ arises because the Mill's ratio uses the Gaussian tail $\bar{\Phi}(x) \approx \phi(x)/x$, whereas the exact overlap distribution on the sphere, $f(q) \propto (1-q^2)^{(N-3)/2}$, has a *lighter* tail than the Gaussian for finite N. Open circles show the corrected values $\langle K \rangle \cdot e^{cN}$ with $c \approx 0.083$, which collapse onto the analytical curves, confirming that the discrepancy is a multiplicative finite-N effect that vanishes as $N \to \infty$. **Panel 3** (bottom-left): Survival probability of a Galton–Watson branching process with Poisson(λ) offspring ($N = 50$), truncated at BFS depths $1, 3, 10$, and 50. The depth-1 curve matches the analytical $P = 1 - e^{-\lambda}$ (dashed), while deeper explorations sharpen the transition into a step function at $\alpha_c \approx 0.305$ (dotted vertical line), analogous to finite-size scaling in standard percolation. **Panel 4** (bottom-right): Mean BFS neighbor counts at depth 1 ($\langle K_1 \rangle$, circles) and depth 2 ($\langle K_2 \rangle$, squares) from exact enumeration on $S^{N-1}(\sqrt{N})$ ($N = 25$, 200 realizations per α). The depth-1 count tracks the analytical λ (dashed), while new depth-2 neighbors remain negligible ($\langle K_2 \rangle \ll \langle K_1 \rangle$), confirming that connectivity is established at depth 1 and the BFS effectively terminates—consistent with the analytical argument that two depth-1 patterns cannot bridge to a common depth-2 pattern when $\varphi_c^2 < \varphi_c$.

6 Implications for the simulation protocol

The percolation picture justifies the Poisson pattern reduction scheme (v5 approach) with an important correction: **the background energy S_{bg} should be removed**.

1. The energy computed from just the $K+1$ retained patterns (target + K active neighbors) is *exact* for all states reachable from the target basin. Patterns outside the connected component are separated by infinite barriers and cannot be reached.

2. The S_{bg} term in v5 artificially smooths the energy landscape, removing the infinite barriers that confine the state. This distorts the dynamics near the phase boundary.

3. With the Poisson scheme (no S_{bg}), memory scales as $\mathcal{O}(N \cdot K_{\max})$ instead of $\mathcal{O}(N \cdot M)$, enabling much larger N for sharper phase boundaries.

4. For $\alpha > \alpha_c$ (supercritical), K grows exponentially but can be capped at some K_{cap} without affecting the measured overlap φ at the target—the state delocalizes among the K_{cap} nearest basins, which is sufficient to show non-retrieval.

Connection to the Hopfield model. In the classical Hopfield model, the coupling structure creates an exponentially large number of spin-glass metastable states via random-matrix frustration. In the LSR model, the log-$\sum$-exp energy function confines the state to discrete basins separated by infinite barriers. The phase transition is purely *geometric*—basin percolation—rather than arising from frustrated interactions. This explains the absence of the spin-glass phase in dense associative memories.

References

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